Uncertainty quantification (step 16) and explainability (step 17)

Two analyses on top of the trained models. Both run **on Kaggle** from `medlift_kaggle.zip` (the RunPod run also runs them once at the end, so their outputs are already in the bundle).

step	question	needs	Kaggle time (estimate)
16 `16_uncertainty.py`	How sure are we about each nodule's volume / mass, and about its clinical category?	`work/results/pipeline.jsonl` (step 9)	seconds, CPU
17 `17_explain.py`	What did the X-rays measure, what did the diffusion prior add, what does the detector look at, which nodules are hard?	both models + the 15 bundled test patients + `pipeline.jsonl`	~30-50 min on a T4 for 12 nodules (part B); parts A and C: minutes

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Step 16: uncertainty quantification with conformal predictive systems (`crepes`)

What it does, in plain words

Step 9 measures every nodule 8 times (8 diffusion samples, the "ensemble"). Their spread is a hint of how unsure the model is, but a hint is not a guarantee. **Conformal prediction** turns it into numbers with a guarantee: we look at how wrong the method actually was on the **cal** patients (truth known) and use exactly that to size the uncertainty for new patients. Step 10 already does this for 90% intervals.

Step 16 uses a **conformal predictive system** (CPS, from the `crepes` library). Instead of one interval it gives each test nodule a whole calibrated distribution: "the true volume is below v with probability p", for every v. From that we get:

1. **Intervals at any level**, in four variants:
   - *standard*: the same relative width for every nodule;
   - *normalised by the ensemble spread*: wider where the 8 samples disagree (this is step 10's method);
   - *Mondrian size groups*: calibrated separately for small / medium / large nodules (by the *estimated* size, known before the truth). Plain conformal intervals are right **on average**; they can be too narrow for small nodules and too wide for large ones. Mondrian makes them right **inside each group**;
   - *Mondrian + normalised*: both.
2. **Calibrated probabilities of clinical decisions.** The European Position Statement on lung cancer screening (EUPS, Oudkerk et al. 2017; built on NELSON volumetry, the categories the demo already shows) manages solid nodules by volume: < 100 mm3 (low risk), 100-300 mm3 (indeterminate, 3-month follow-up),

> = 300 mm3 (refer). For each nodule: P(volume >= 100 mm3) and P(volume >= 300 mm3), the probability of each category, and the **set of categories** that contains the true one 90% of the time (e.g. "100-300 or >= 300 mm3" when the nodule is close to a threshold). Compared with the uncalibrated alternatives: the share of the 8 samples above the threshold ("ensemble vote"), the spread read as a log-normal error, and the point estimate alone.

Outputs (`outputs/16_uncertainty/`) and how to read them

file	what to look at
`table_uq_intervals.md`	coverage close to 90% (the guarantee); **width**: if "normalised" is narrower than "standard" at the same coverage, the diffusion ensemble's spread carries real information about the error. CRPS: quality of the whole distribution (lower = better). The last row reproduces step 10 with our own code (cross-check: should match "normalised")
`table_uq_groups.md`, `group_coverage.png`	coverage per size group: does plain conformal under-cover small nodules, and does Mondrian fix it?
`table_uq_risk.md`, `reliability.png`	Brier score and calibration error (ECE) of P(V >= 100), P(V >= 300): lower is better; calibrated probabilities should lie on the diagonal of the reliability plot. Category sets: how often they contain the true category, and how often they are a single category
`table_uq_methods.md`	the same for every method (ours vs physics-only)
`uq_test_nodules.csv`	per test nodule: estimate, 90% interval, P(>= 100), P(>= 300), category set, truth

The uncertainty plots (`{volume,mass}_*.png`, all eight in `{volume,mass}_uq_overview.png`)

Test nodules, our method, conformal intervals normalised by the ensemble spread (the pre-specified main variant). Residuals are in log space, log(truth / estimate): 0.1 means the estimate is about 10% too low, -0.1 about 10% too high (errors of volumes and masses are multiplicative).

plot	file	what it shows	what you want to see
conformal predictive distribution	`_cpd_examples.png`, `_cpd_median_nodule.png`	for one nodule, the calibrated probability that the true value is below v, for every v (EUPS thresholds in red for volume)	a steep curve (sure) around the truth; flat = unsure
prediction intervals	`_intervals.png`	every test nodule's 90% interval (bar), estimate (dash) and truth (dot), sorted by the estimate	~90% green dots; bars wider where nodules are hard
regression with uncertainty	`_regression.png`	estimate vs truth, each interval as a vertical bar	points near the diagonal; a bar that crosses the diagonal = covered
residuals	`_residuals.png`	residual vs estimate, with each interval in the same units; grey = cal nodules	no trend with size (else size-dependent bias); red points = misses
distribution of residuals	`_residual_hist.png`	histogram of cal and test residuals, the normal curve a Gaussian method would assume, the cal 5% / 95% quantiles	centred near 0 (mean = bias); cal and test alike (exchangeable); skew or heavy tails are fine for conformal

plot	file	what it shows	what you want to see
Q-Q of residuals	`_qq.png`	residuals (and residuals / ensemble spread, the conformal scores) against normal quantiles	points off the line in the tails = not Gaussian: this is why distribution-free conformal intervals are used instead of +-1.645 sd
distribution of interval widths	`_interval_widths.png`	relative width of the 90% intervals per variant	standard: one width for all; normalised / Mondrian: spread out; narrower at the same coverage = more informative
coverage at every level	`_coverage_levels.png`	observed coverage at 50-95% nominal, for conformal (standard, normalised) and the raw ensemble spread read as +-z sd	conformal on the diagonal; the uncalibrated spread usually below it (over-confident)

`reliability.png` (calibration of P(V >= 100 / 300 mm3)) and `group_coverage.png` (coverage per size group) complete the set.

In the demo: every report now shows the calibrated category probabilities, e.g. `EUPS category, calibrated probability (step 16): < 100 mm3: 4% | 100-300 mm3: 71% | >= 300 mm3: 25%` (read from `outputs/16_uncertainty/cps_calibration.json`; computed with plain numpy, tested to be identical to `crepes`). This is what a doctor can act on: not only "about 180 mm3" but "71% likely indeterminate, 25% likely refer".

Technical choices (fixed in advance, not tuned on the test patients): log space (errors are multiplicative); sigma = ensemble log-std + 0.02; 3 equal-count size groups on the cal nodules (fewer if a group would have < 19 nodules, the minimum for a finite 90% interval); probabilities from the pooled normalised CPS (all cal nodules -> finest resolution), the Mondrian version shown as a second row; the CPS value is the expected smoothed conformal p-value (tau = 1/2): deterministic and valid on average.

Install on Kaggle: `pip install crepes scikit-learn` (scikit-learn is usually preinstalled).

Step 17: explainability

A. Error drivers (CPU, seconds)

Which nodule properties go with large volume / mass errors of our method: size, solidity (ground glass -> solid), density, the radiologists' own disagreement, malignancy rating, the ensemble spread, the physics data misfit. Spearman correlation with a bootstrap 95% interval, plus the error per size and solidity group. Output: `table_error_drivers.md`. Read: a correlation whose interval excludes 0 is a real pattern ("small ground-glass nodules are hardest"); this tells a doctor when to trust the number less.

B. Seen / unseen split: what the X-rays measured vs what the prior filled in (the main part)

The idea. Fifteen X-ray views from a 30 deg sweep cannot see everything in 3D. Every 3D volume can be split exactly into two parts:

- **seen** = the part the 15 views measure; it is fully determined by the X-ray data;

- **unseen** = the part the views cannot see at all: adding or removing it does not change a single measured pixel. Mathematically: the null space of the projection operator.

A physics-only reconstruction (SIRT) contains only the seen part. Whatever unseen content our diffusion reconstruction contains was supplied by the **learned prior**: the data cannot have put it there. We compute the split for each nodule box (40 x 50 x 40 mm) of four volumes: the **true CT**, ROI-SIRT, diffusion, diffusion + Gaussians, using the exact projector of the simulation (CGLS, checked by |A unseen| / |A x| close to 0).

What it shows (`table_explain.md`, `<nodule>_seen_unseen.png`):

number	meaning
unseen share of the **true** nodule's signal	how much of a real nodule is invisible to 15-view DTS: the part no physics method can ever recover
unseen share of the **true** nodule's **mass**	close to 0: the mass is seen by every view. **This is why H1 holds** (mass error < volume error), now shown on each nodule, not only as a formula
volume from the seen part only / full	how much the depth smear inflates the volume when the unseen part is missing
unseen share of **ours**	how much the prior added
unseen skill vs truth	1 = the prior filled in exactly the true invisible part; 0 = no better than adding nothing (physics); **< 0 = the prior invented structure** (hallucination). This is the honest test of the generative prior
seen skill vs truth	how well the measured part matches the truth (noise, ROI-SIRT data consistency)
Spearman(unseen share of the truth, our volume error)	do less visible nodules get larger errors?

The figure shows a side view (depth -> horizontal, the direction DTS resolves worst): the true nodule is a compact blob; its seen part is smeared in depth; its unseen part is the sharp core with negative side lobes -- exactly what the prior has to supply. Below: depth profiles through the nodule centre, and the disagreement of the diffusion members (where the model itself is unsure).

C. Detector saliency

For each suggestion of the learned detector on the test scans: which voxels of the DTS volume made it fire (|input x gradient| of the heatmap at the suggestion, 64^3 window). `table_saliency.md`: share of the saliency within 10 mm of the suggestion for true vs false positives, and on the nodule itself for true positives. `saliency_examples.png`: 3 true and 3 false positives (coronal and side view). Read: a detector that "looks at" the nodule concentrates its saliency there; false positives that look at vessels or ribs explain the typical mistakes.

Run on Kaggle

After Cells 1-2 of the Kaggle notebook (RUNPOD_FULL_RUN.md section 10 / README Part B):

```
        !pip install -q crepes scikit-learn
        !python scripts/16_uncertainty.py
        !python scripts/17_explain.py --limit 12 --members 4 --outputs /kaggle/working/outputs_kaggle
        !cat outputs/16_uncertainty/table_uq_*.md
/kaggle/working/outputs_kaggle/17_explain/table_*.md
```

The bundle already holds step 17 from RunPod (30 nodules, 8 members): `--outputs` keeps the Kaggle run (12 nodules of the 15 bundled patients) separate. `--skip-maps` runs only parts A and C (minutes).

Does this add novelty? (honest answer, literature checked Oct 2026)

addition	what already exists	what is new here
Conformal predictive distributions for nodule volume / mass with **calibrated EUPS category probabilities** and category sets	conformal *intervals* for CT measurements; conformal risk control for nodule *detection* (Hulsman et al., MLR Press 2025); CPS in general (Vovk et al.; `crepes`, Boström 2022)	decision-level calibrated uncertainty for DTS nodule volumetry; test of whether the generative ensemble's spread is informative (normalised vs standard) and of size-conditional coverage (Mondrian)
Seen / unseen (range / null space) analysis of the reconstruction	null-space "hallucination maps" (Bhadra, Kelkar, Brooks, Anastasio, IEEE TMI 2021); range-null-space *reconstruction* methods (DDNM, RN-SDEs 2024)	used as an **explanation of a clinical measurement** in DTS: per-nodule proof that the mass is seen and the depth shape is not (H1), how much of a real nodule is invisible to 15-view DTS, and an "unseen skill" score of the diffusion prior against the true CT
Detector saliency, error drivers	standard XAI	supporting analysis, not a claim

What you can claim: calibrated, decision-level uncertainty for DTS nodule measurement; a physics-grounded explanation of what the X-rays determine and what the learned prior contributes, quantified against the truth. **Do not claim** the first use of conformal prediction in lung nodules, the first null-space hallucination analysis, or that saliency maps are new; cite Bhadra et al. 2021 for the null-space idea.

What makes it a contribution and not decoration: each part answers a question the main results raise -- *Can a doctor act on the number?* (category probabilities), *Why is mass better than volume?* (mass is seen), *Is the diffusion model inventing the nodule?* (unseen skill), *When should I not trust it?* (error drivers, size-group coverage). Report the numbers whichever way they come out; a negative unseen skill would be an important, honest finding about generative priors in DTS.

Limitations

- The seen / unseen split is local (nodule box, the rest of the prism treated as known) and uses the simulation's own projector; in a real scanner the operator is only approximately known.
- Finite iterations: components the data barely constrain count as unseen (practically invisible at this noise level); the reported residual quantifies it.
- Category probabilities use the radiologist consensus volume as truth, which itself varies between readers (step 10 reports the reader CV). The EUPS thresholds are for solid nodules; we apply them to all for illustration.
- With ~75 cal / ~75 test nodules, group coverages and reliability curves are coarse (a few nodules per bin).